

Jawut Object Annotation Tools — User Guide

Draw boxes on images, manage your classes, export a YOLO dataset.

This guide walks through the whole job from a blank screen to a training-ready dataset. The screenshots are from a real session on real site photographs.

Before you begin

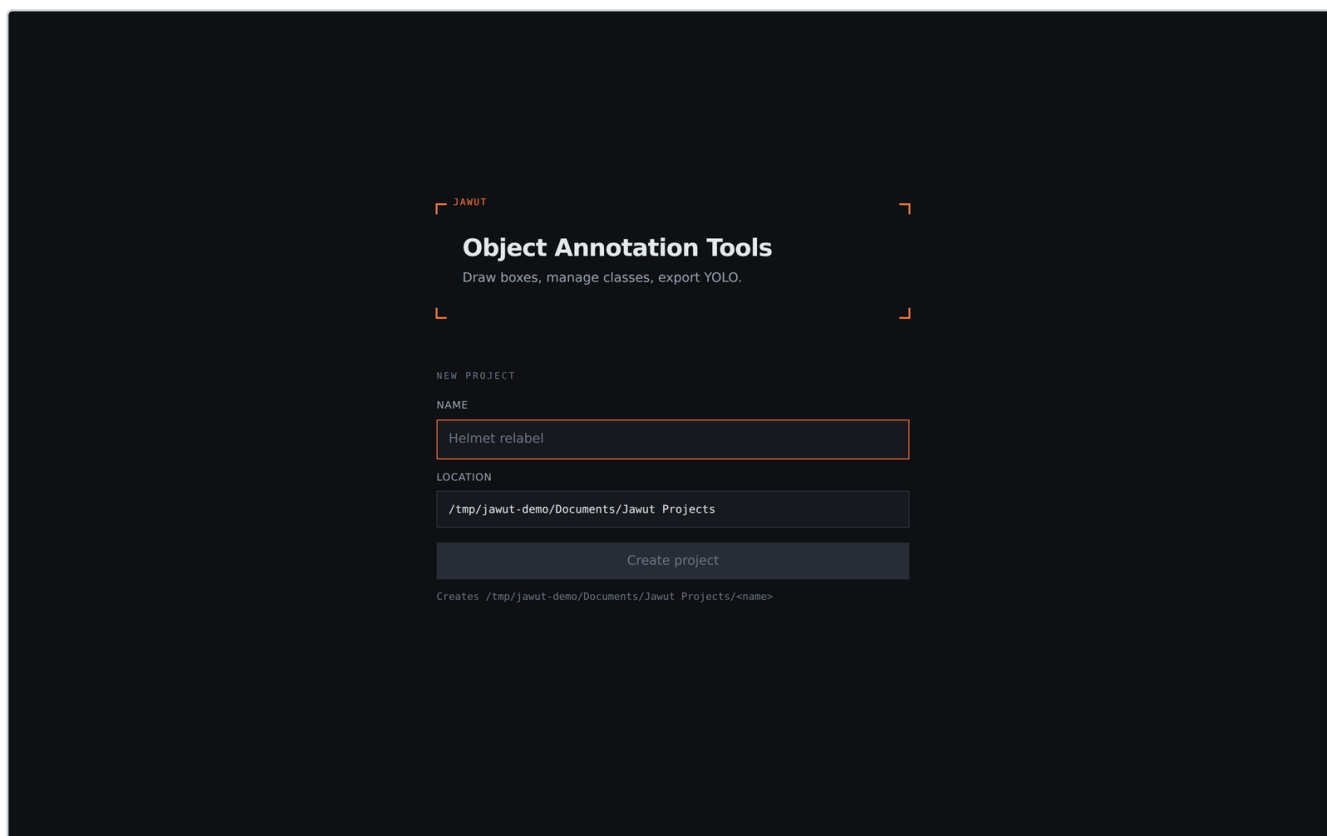
Download the zip from the project's Releases page, extract it anywhere, and run **Jawut Object Annotation Tools.exe**.

Nothing else is needed. No Python, no database, no installer. The whole application is inside that folder.

If Windows shows "Windows protected your PC" — click **More info**, then **Run anyway**. This appears because the application is not code signed, not because anything is wrong with it.

Step 1 — The welcome screen

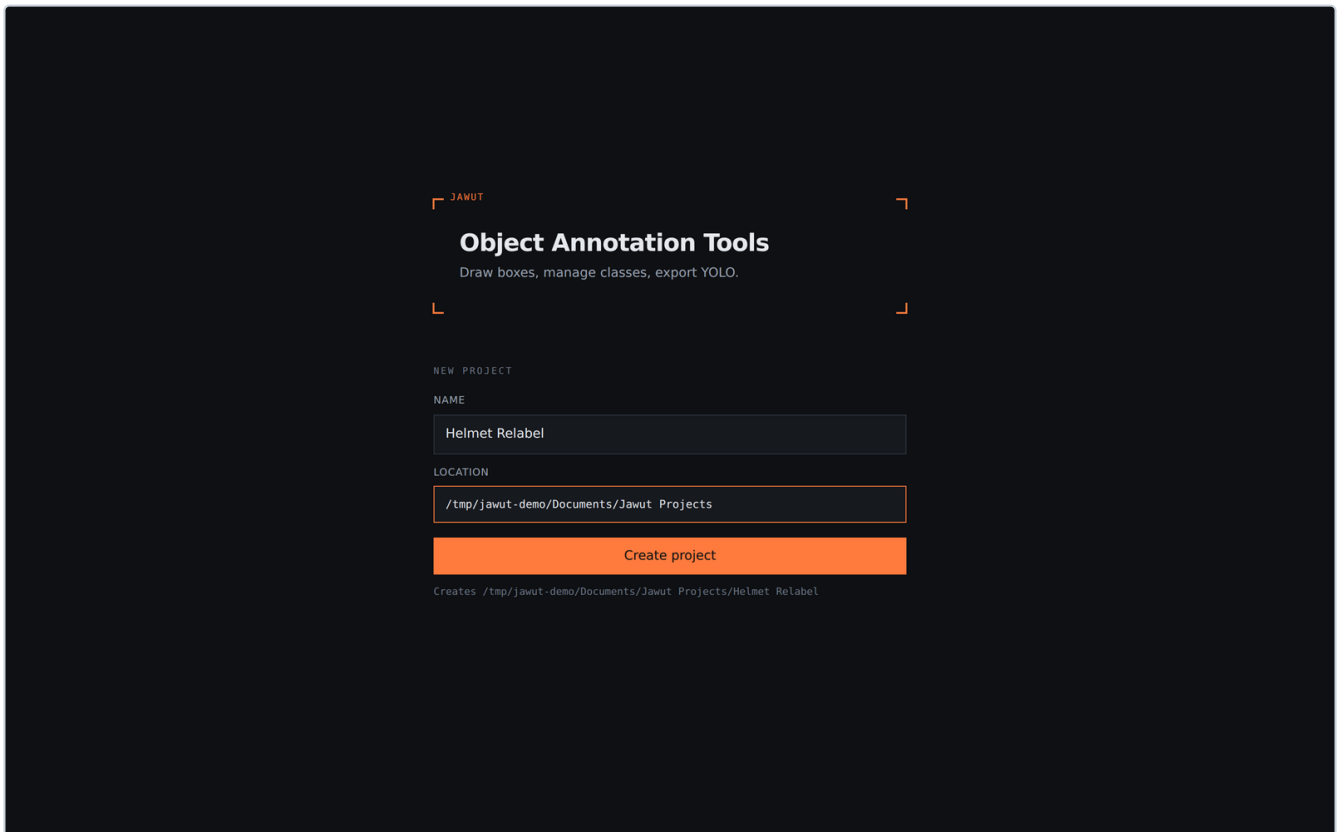
When the application opens for the first time you see this. There is nothing here yet because you have not made a project.



A **project** is one labeling job: a set of images, the classes you are using, and every box you draw. You can have as many as you like, and they never interfere with each other.

Step 2 — Make a project

Type a name, check where it will be saved, and click **Create project**.

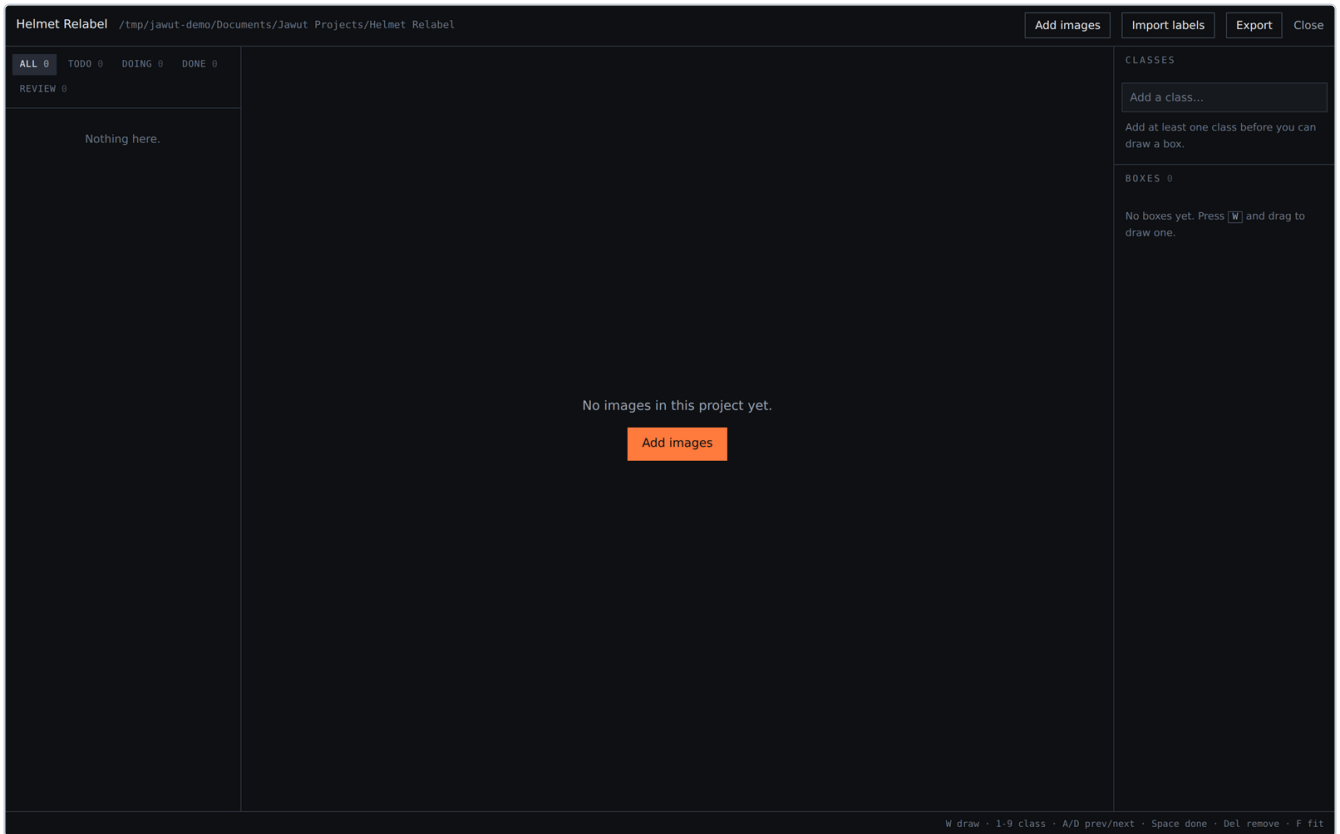


The grey line under the button shows exactly which folder will be created, so there is no guessing.

Where your work lives. A project is a plain folder containing a `project.db` file and a copy of your images. To back it up, copy the folder. To move it to another computer, copy the folder. There is nothing else to keep track of.

Step 3 — The workspace

The project opens. It is empty, which is expected — there are no images yet.

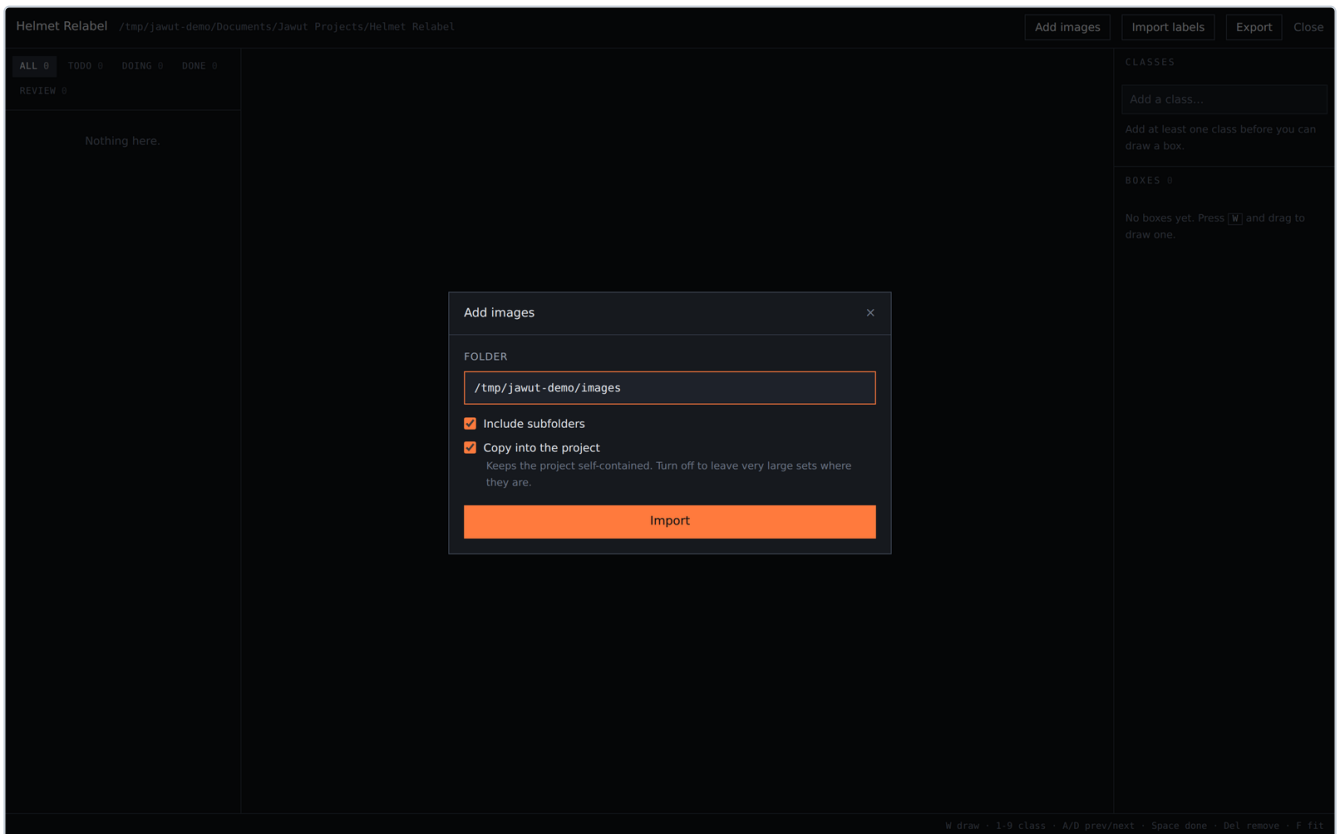


The layout stays the same throughout:

Area	What it holds
Left	Every image, with a coloured dot for its state, and filters along the top
Middle	The image you are working on
Right	Your classes on top, the boxes on the current image below
Bottom	The current file, its size, its state, and the keyboard shortcuts

Step 4 — Add your images

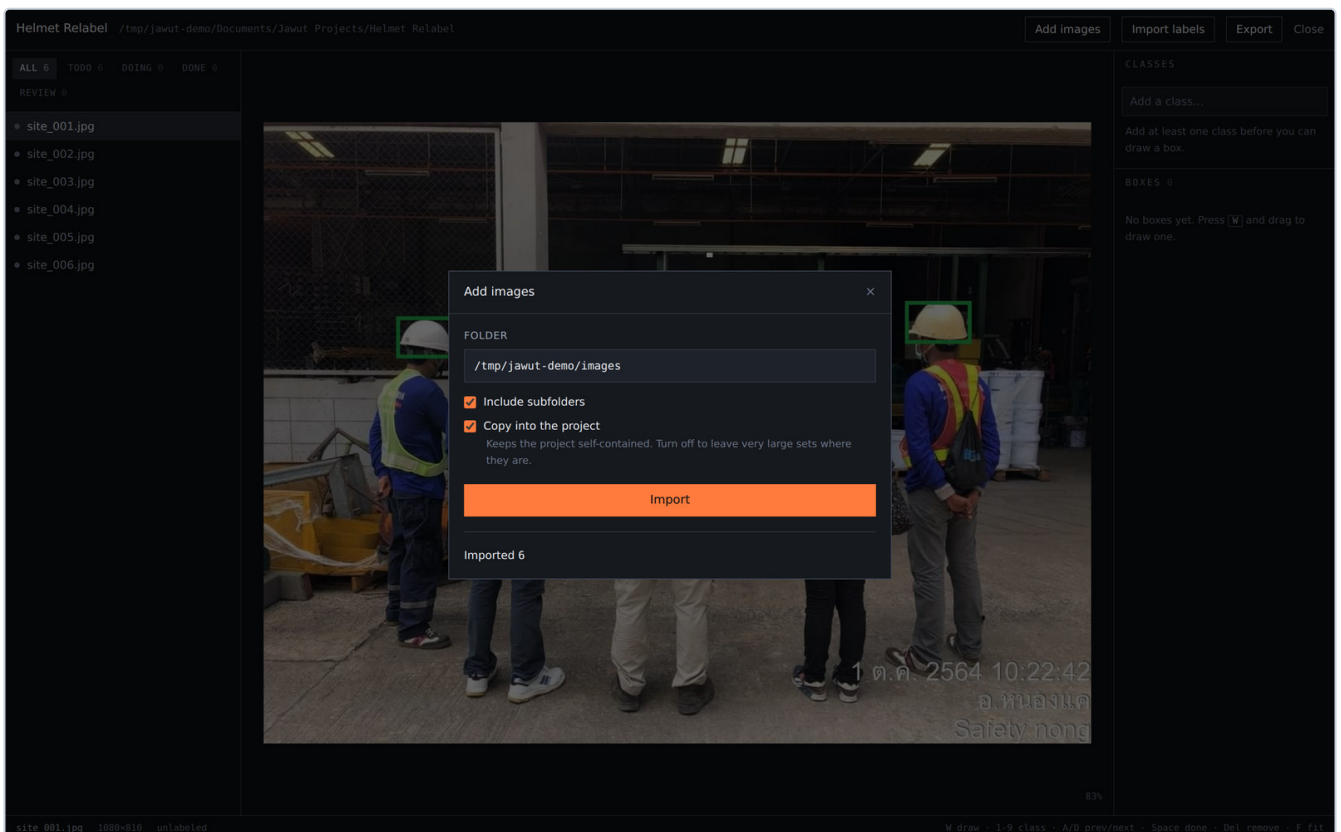
Click **Add images**, then give it the folder your photographs are in.



Two options are worth understanding:

- **Include subfolders** — on by default. Finds images in folders inside the folder you chose.
- **Copy into the project** — on by default, and worth leaving on. Your photographs are copied into the project, so the project folder is self-contained. Moving or reorganising your original photo folders later cannot break it. Turn it off only for very large sets you do not want to duplicate on disk.

Click **Import**. You get a report:

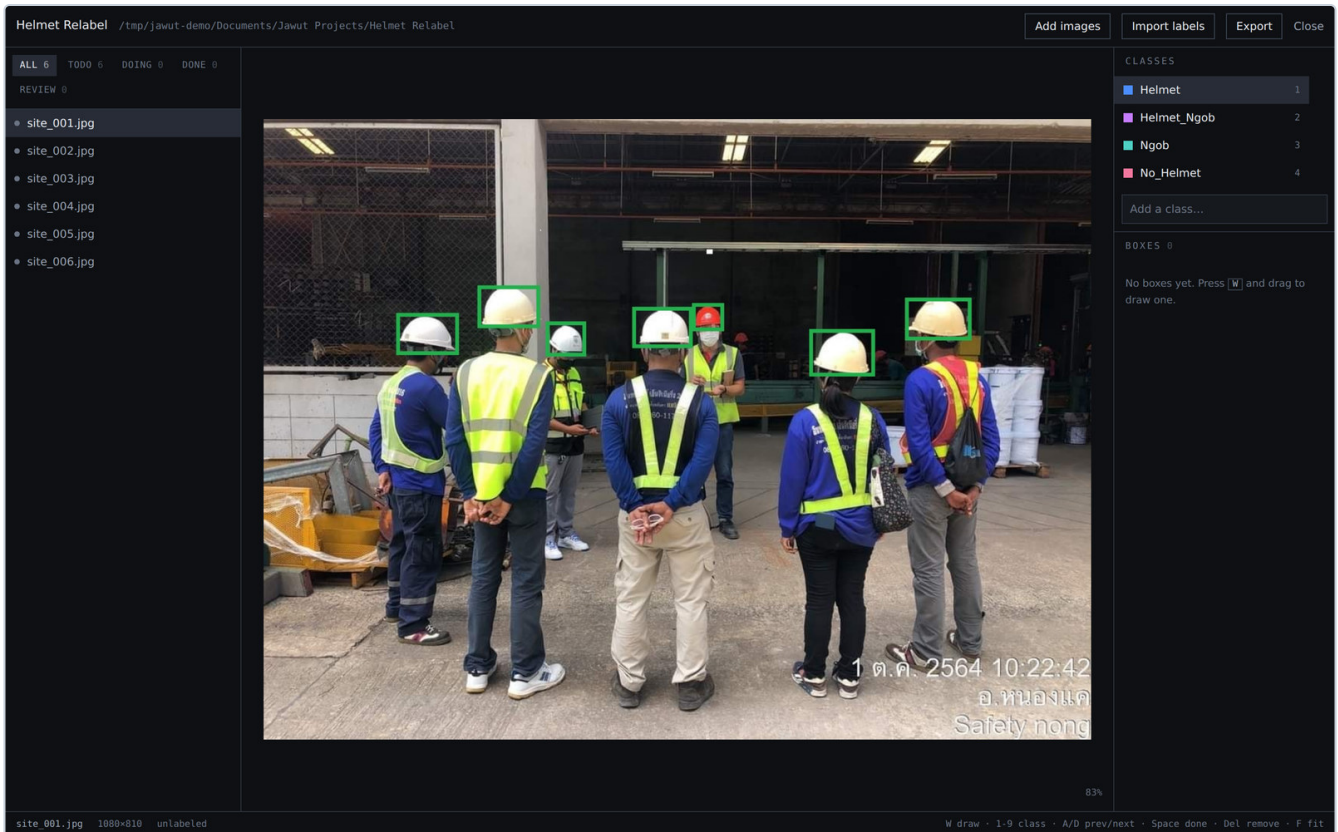


Read this before moving on. It tells you how many images came in, how many were already in the project, and how many were **skipped** — each with the reason. A skipped image is usually a damaged file, which is much better to know now than halfway through labeling.

The same photograph cannot be imported twice. The application compares file contents, not names, so importing the same folder again adds nothing.

Step 5 — Set up your classes

A class is a category you are labeling — a kind of object. Type a name in **Add a class...** on the right and press Enter. Repeat for each one.



Each class gets a colour and a number. **The number is its keyboard shortcut.** The first class is **1**, the second is **2**, and so on.

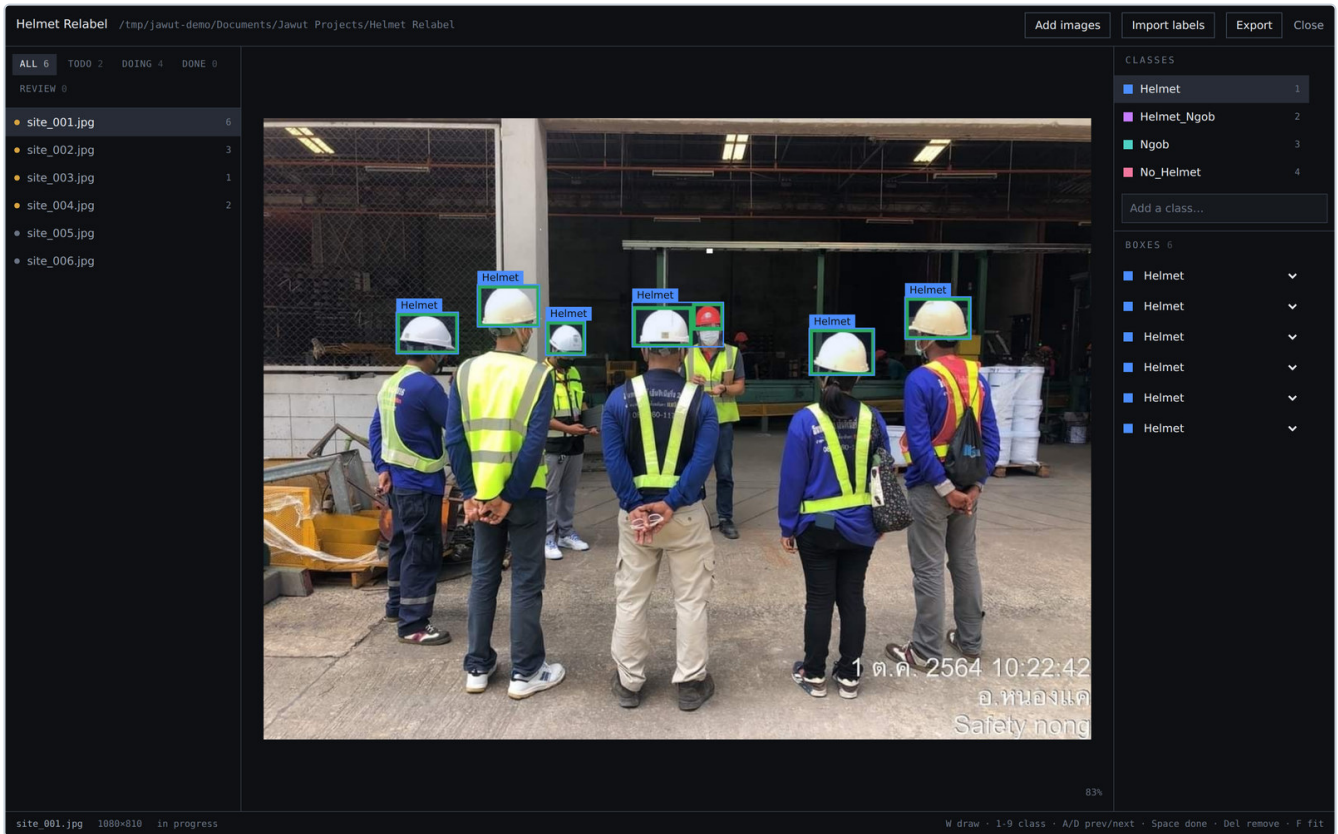
The order matters at export time, because it becomes the class index in the exported dataset. It does not matter while you are labeling, and you can change it later without breaking anything.

To rename a class, double-click it.

Step 6 — Label

Select an image on the left, then:

1. Press **W**
2. Drag a box around the object
3. Press the **number key** for its class



Every box you draw appears in the list on the right, and the number beside the filename on the left counts them.

The keyboard is the whole job

Labeling a thousand images with the mouse alone is slow. These are worth learning — they are also listed along the bottom of the window at all times.

Key	What it does
W	Start drawing a new box
1 – 9	Set the class of the selected box
A / D	Previous / next image
Space	Mark this image finished and go to the next one
Delete	Delete the selected box
F	Fit the image to the window
Esc	Cancel drawing, or deselect

Adjusting a box

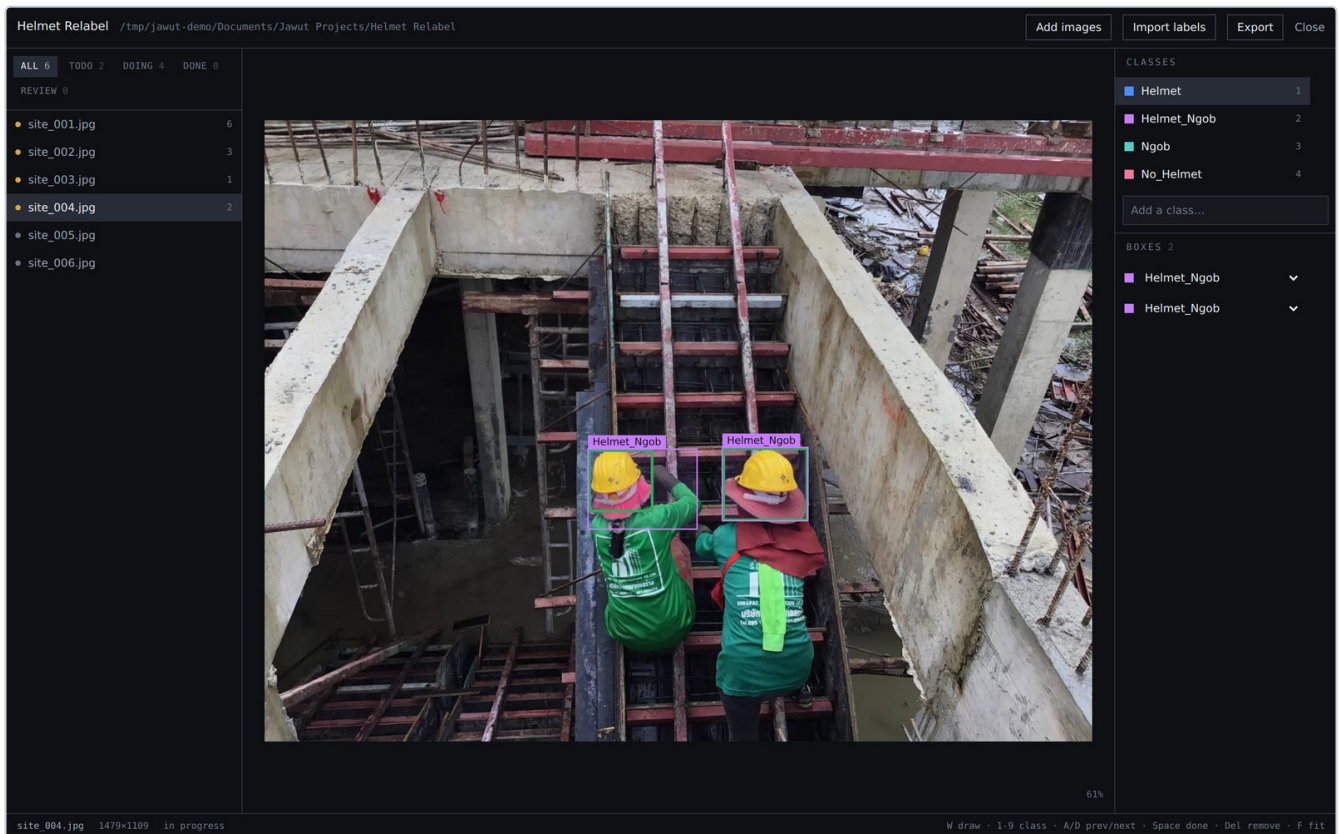
Click a box to select it. Eight small square handles appear on its edges and corners — drag any of them to resize, or drag the middle of the box to move it.

Scroll the wheel to zoom, and drag the background to pan. Zooming in is worth doing when a boundary is unclear; the image stays sharp rather than going blurry.

Changing the class of a box

Two ways, both instant:

- Select the box and press its number key.
- Use the dropdown next to the box in the right-hand list.



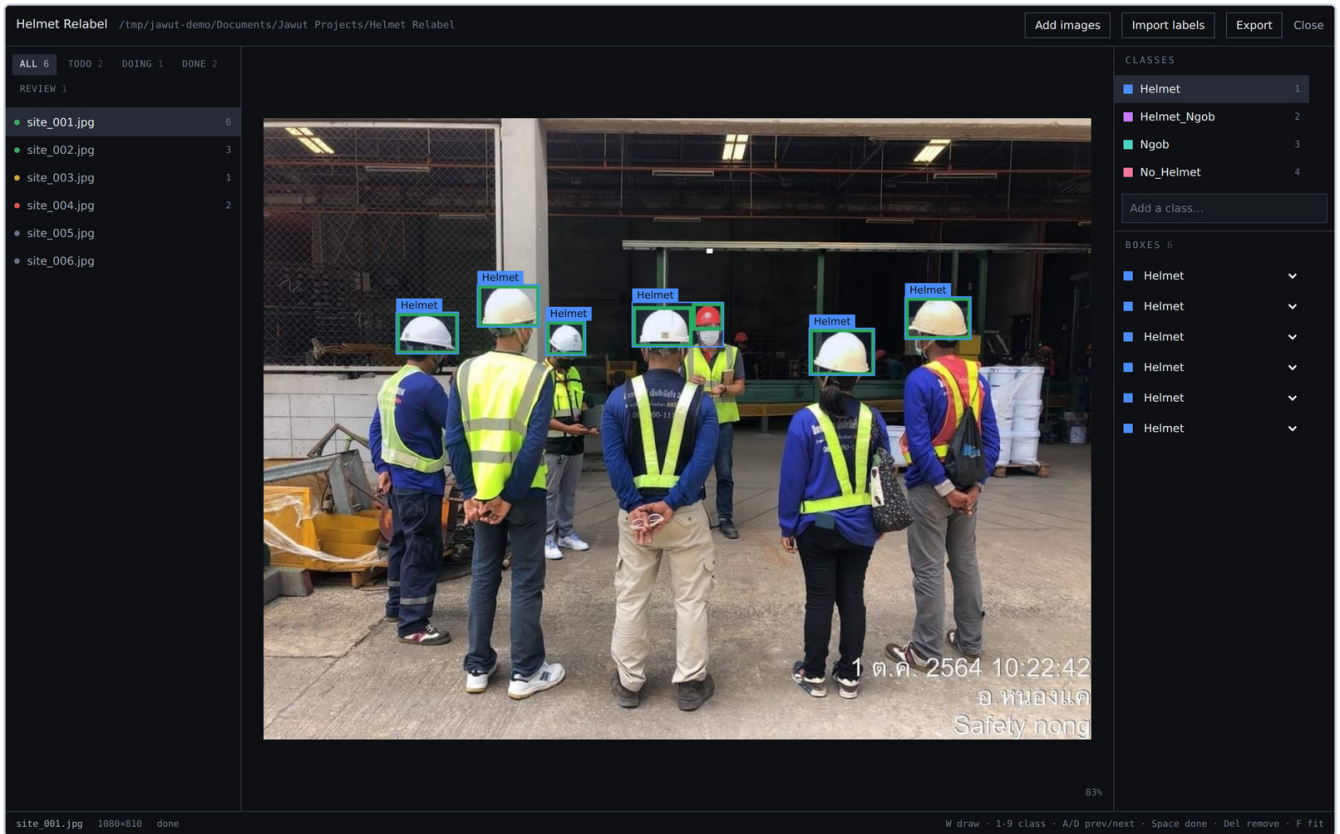
Images with nothing in them

If an image contains none of your classes, **press Space without drawing anything**. That records "I looked at this, there is nothing here", which is real and useful information — those become valid background images in the export.

This is different from never opening the image at all, and the two are exported differently. Skipping an image is not the same as clearing it.

Step 7 — Track your progress

Every image is in one of four states, shown as a coloured dot and counted in the filter buttons at the top left.



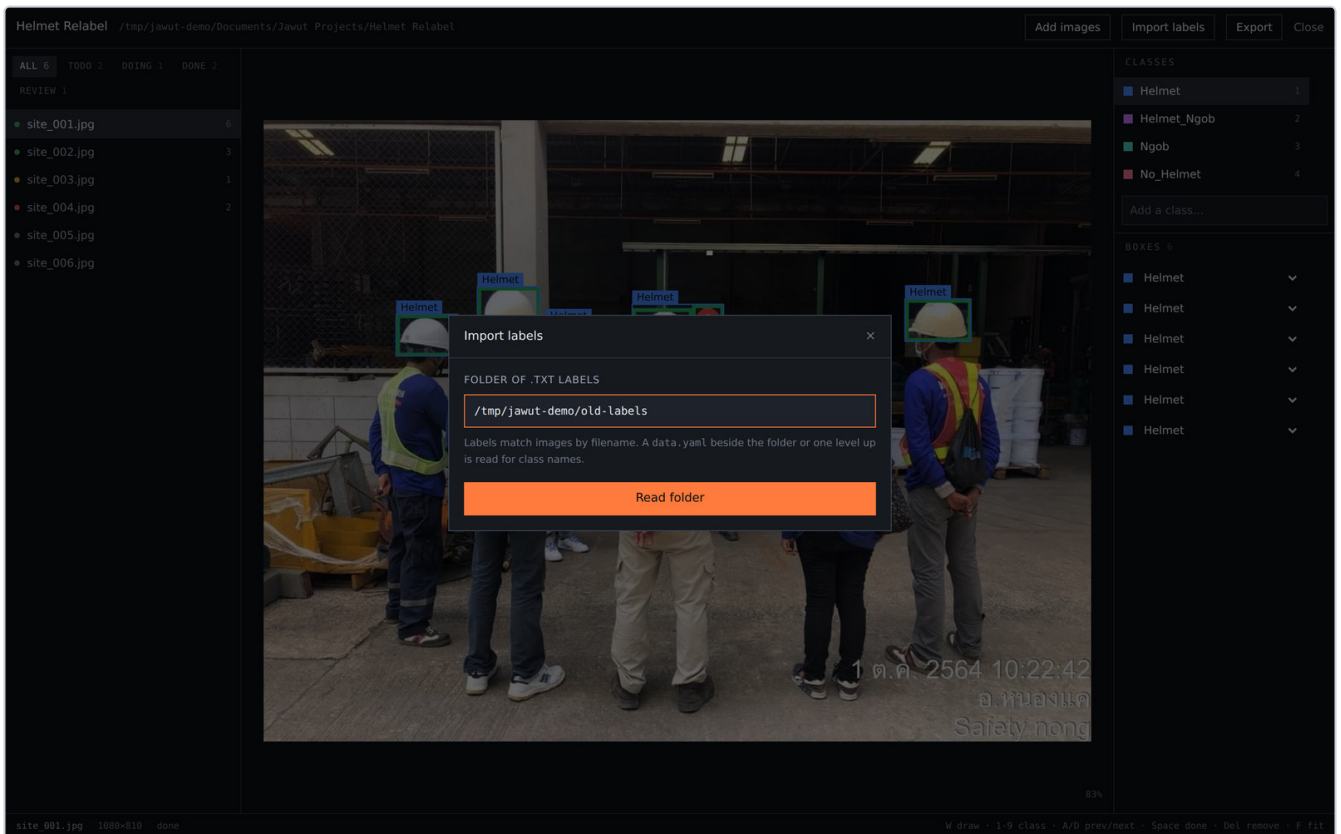
Filter	Dot	Meaning
Todo	grey	Nobody has opened this image yet
Doing	amber	It has boxes, but has not been marked finished
Done	green	Finished — either boxed, or confirmed empty
Review	red	Flagged, or something the importer could not read

Click a filter to show only those images. The arrow keys then move only within that set, so filtering to **Todo** and pressing **D** walks you through exactly the work that is left.

Step 8 — Bringing in labels you already have

If you already have YOLO `.txt` labels — from a previous tool, or a colleague, or an earlier attempt — you do not have to start over.

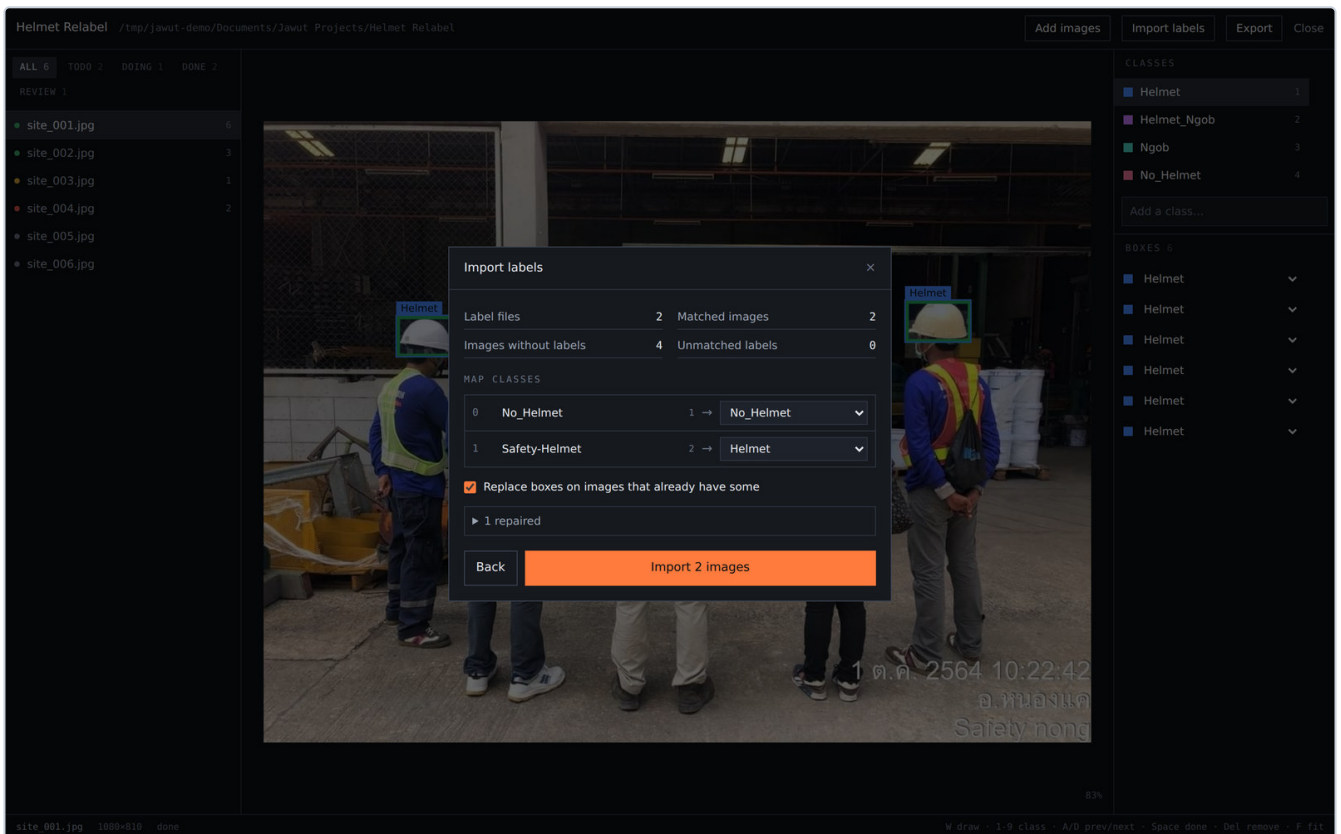
Click **Import labels** and point at the folder holding the `.txt` files.



Labels are matched to images **by filename**: `site_004.txt` belongs to `site_004.jpg`. If a `data.yaml` sits beside the folder or one level above it, the class names are read from it automatically.

Click **Read folder**. Nothing has been changed yet — this only reads and reports.

Mapping the classes



This is the important part. The old labels have their own class names, and you decide where each one lands. For each incoming class you can:

- **Map it to one of your classes** — the count beside the name tells you how many boxes will move.
- **Create it as a new class** — the default when the name does not already exist.
- **Ignore it** — those boxes are not imported.

Two old classes can map to the same new class, which is how you merge categories.

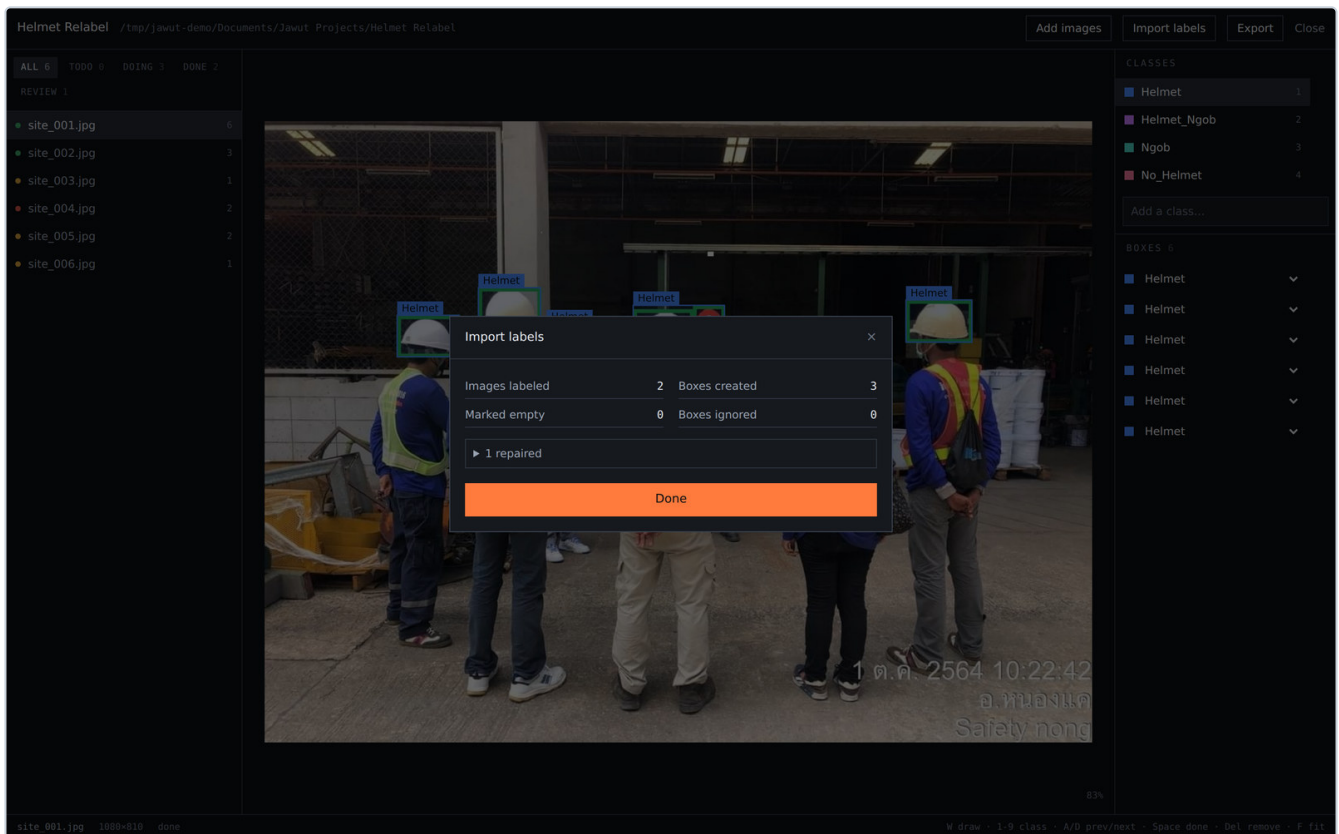
The summary above shows how many label files were found, how many matched an image, and how many labels had no matching image at all.

Click the expander at the bottom to see every issue the importer found, each with its file and line number. It reports two kinds:

- **Repaired** — something it could fix unambiguously and did. Coordinates given in pixels instead of 0-1, values a hair outside the valid range, or extra data from a keypoint-labeling tool where the box itself is still perfectly good.
- **Rejected** — something it will not guess at. A line it cannot read, a class that is not in your mapping, or a box with no area.

Nothing is ever discarded silently. If a line did not become a box, it is in that list with the reason.

The result

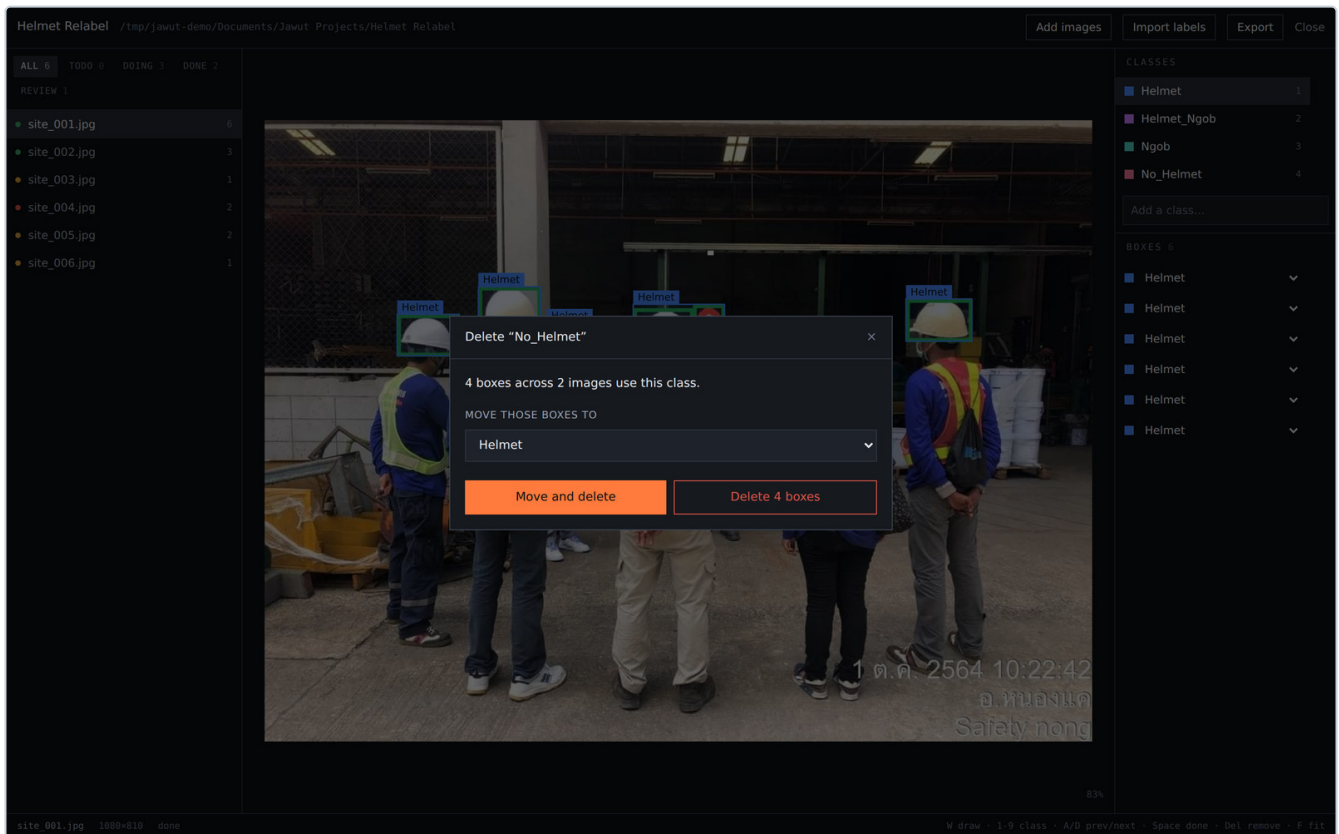


Images that received boxes become **Doing**. Images whose label file was empty become **Done** — an empty label file means "reviewed, nothing here". Images whose labels could not be read are marked **Review** so you can find them. Images with no label file at all are left untouched.

Step 9 — Changing your mind about classes

You can rename, reorder and delete classes at any point, including after you have labeled thousands of images. Nothing breaks.

To delete a class, hover over it and click the **x**.



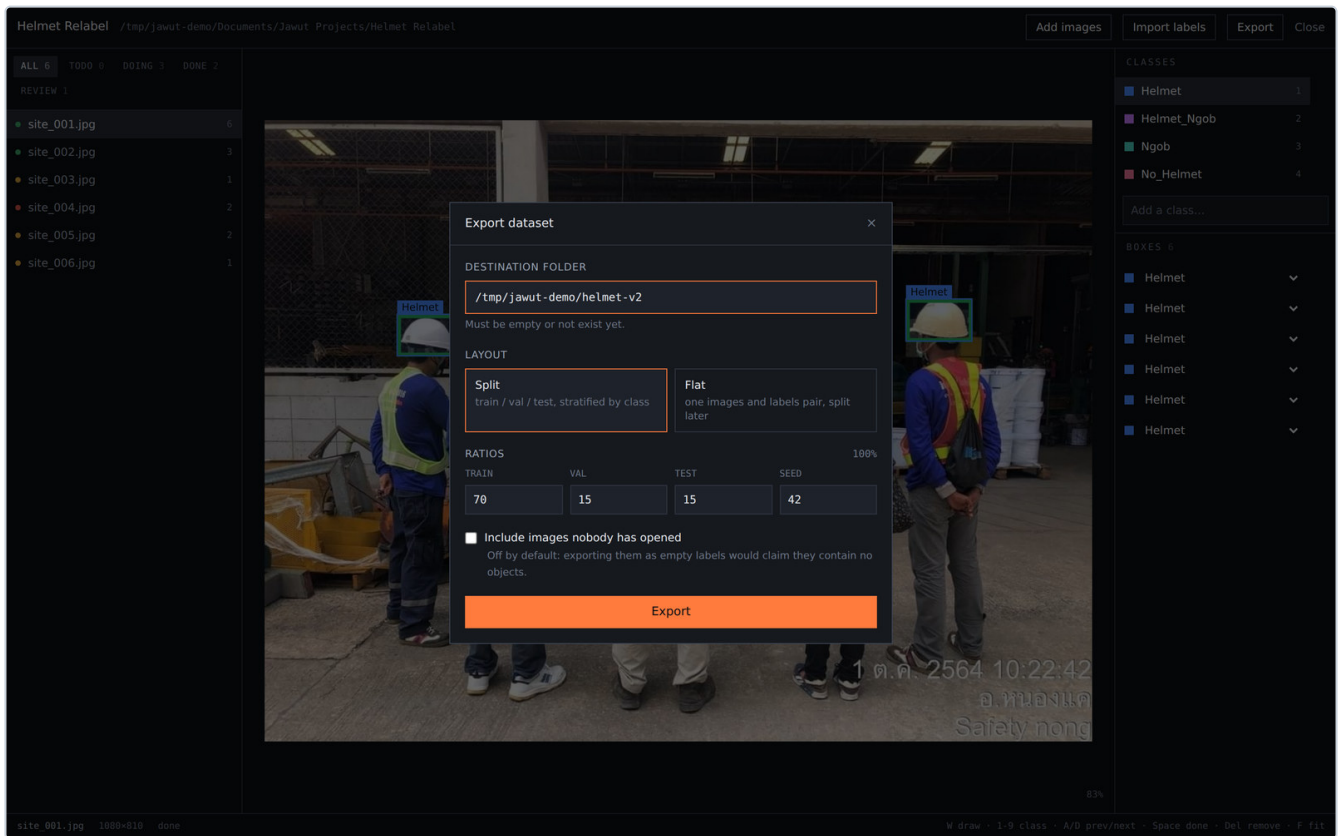
The dialog tells you exactly what it would cost — how many boxes, across how many images — before you commit. Then you choose:

- **Move and delete** — the boxes are reassigned to another class and kept. Use this when you are merging two categories.
- **Delete the boxes** — the boxes go too. Use this when the category was a mistake.

Either way it happens in one go. It cannot half-finish and leave your project in a strange state.

Step 10 — Export

Click **Export**, and choose an empty folder to write to.

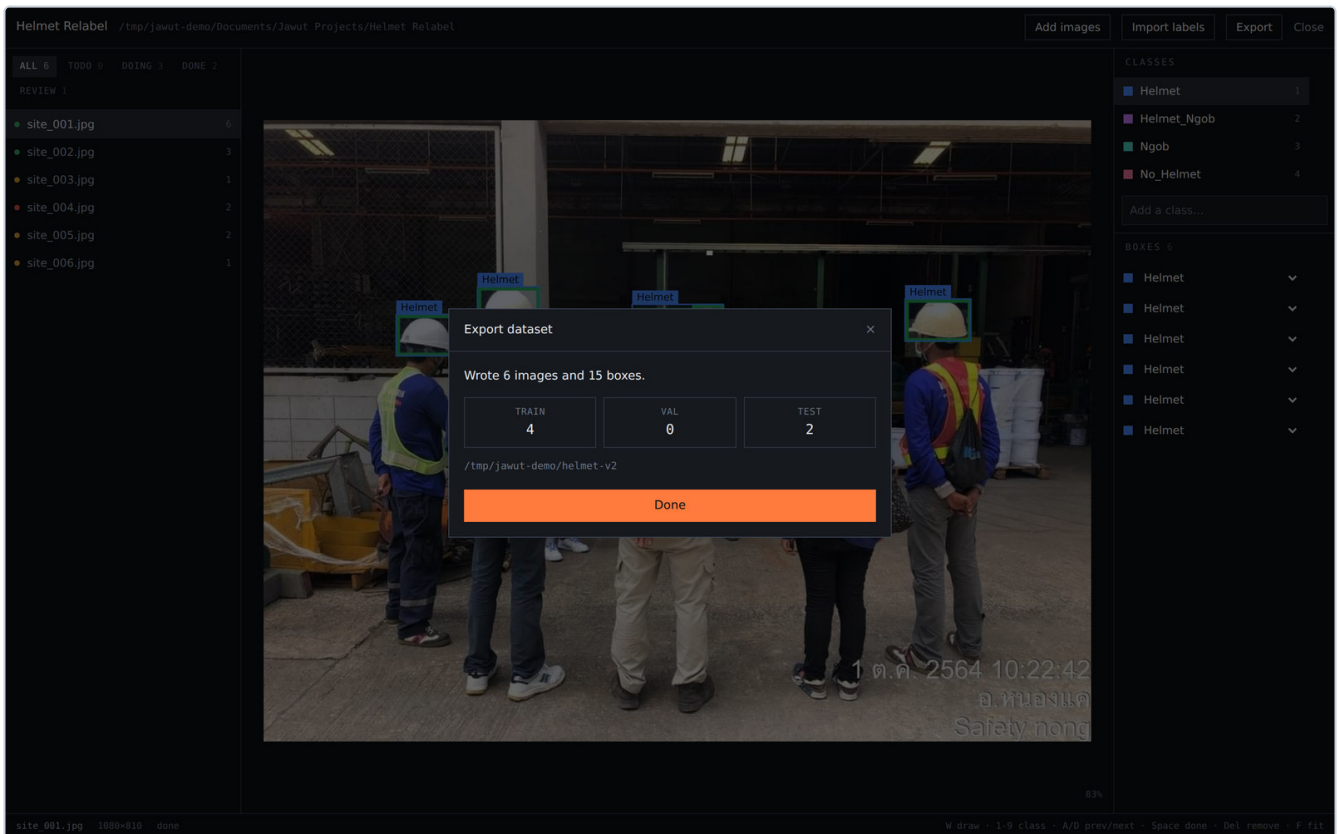


Layout — two choices:

- **Split** writes `train`, `val` and `test` folders, ready to train on directly. The default 70/15/15 is a reasonable starting point. The split is stratified, which means a class you have very few examples of is spread across the splits rather than landing entirely in one — otherwise its validation numbers would mean nothing.
- **Flat** writes one `images` and `labels` pair with no split, for when you would rather divide it yourself later.

Seed — the same seed always produces the same split. Leave it alone unless you have a reason, and write it down if you do.

Include images nobody has opened — leave this off. Exporting an untouched image would claim it contains no objects, which nobody has actually checked.



What you get

```
data.yaml           class names and where the splits are
train/images       train/labels
val/images         val/labels
test/images        test/labels
export_manifest.json  seed, ratios, and which image went where
```

Each label file holds one line per box: the class number, then the centre, width and height of the box as fractions between 0 and 1. Standard YOLO — any training script will read it.

Keep `export_manifest.json`. It records exactly how this dataset was built, which is what lets you reproduce or check it months later.

Where everything is kept

What	Where
The application	Wherever you extracted the zip
Settings and your recent projects	%APPDATA%\Jawut\
Your projects	The folder you chose, by default Documents\Jawut Projects\

If something goes wrong

An image will not display. If it was imported with Copy into the project turned off and the original file has since moved or been deleted, it cannot be found. Import it again from its new location.

The export refuses to run. The destination folder has to be empty. Writing into a folder that already holds a dataset would partly overwrite it, and you would only find out when training gave strange results.

A class count looks wrong after importing labels. Check the mapping you chose. You can import the same folder again with Replace boxes on images that already have some turned on, which redoes that part cleanly.

Your class counts look lopsided in the export. Open `export_manifest.json` and look at the per-image list. A class with almost nothing in it usually means a mapping mistake rather than a labeling one.

The short version

1. **New Project**, give it a name
2. **Add images**, point at your folder
3. Type your class names on the right
4. `W`, drag, number key. `Space` for the next one
5. **Export**, choose an empty folder, **Split**